

Unit 8: Working with heterogenous datasets and a view towards machine learning

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1 Unit 8: Working with heterogenous datasets and a view towards machine learning

In this unit we learn how to deal with heterogenous datasets using data frames. We then explore basic concepts of machine learning. A deep dive into deep learning at UQ is in the [STAT3007 course](#). Theory of machine learning (and statistical learning) is also studied in [STAT3006](#). Other aspects of applied statistics and data analysis are studied in [STAT3500](#). Our purpose with this unit is just to touch the tip of the iceberg on issues of data analysis and machine learning. The Julia language knowledge acquired in the previous units should help.

2 Databases

This course doesn't touch databases, a rich topic of its own. (Do not confuse the term database with data structures or data frames). A database is a system that stores information in an organized and flexible manner. Many databases are **relational databases** and this means that they are comprised of multiple data tables that can be used together. The most common language for dealing with such databases is [SQL](#). It is not a general programming language but rather a language for querying, modifying, and managing databases. At UQ you can study more about databases in the [INFS2200 course](#) as well as several other more advanced courses.

We now show you an example of database and a few things you may expect from SQL.

2.1 Relations

A “relation” is a mathematical term for a set over *tuples* (or *named tuples*), and is a generalization of a function.

For example, consider the function $y = f(x)$. (The *names* here are x and y). An equivalent relation f can be constructed as a set of tuples of the form (x, y) . A particular tuple (x, y) is in the relation f if and only if $y = f(x)$.

Other things which are not strictly functions may be relations. For example, we can construct a relation of real numbers (x, y) where $x = y^2$. Or, equivalently, $y = \pm\sqrt{x}$. Positive values of x correspond to two values of y (positive or negative square root). When x equals zero, there is only one solution for y (also zero). And for negative x there are no real y that satisfy the relation. Note that I say “relation” and not “function” here - functions are only a subset of relations.

Just like functions may have multiple inputs ($x = f(a, b, c)$), a relation may consist of tuples of any size (including zero!). And both functions and relations can exist over finite, discrete sets.

There are precisely two relations over tuples of size zero (i.e. $()$) - the empty relation, and the relation containing the empty tuple $()$. You can use this fact to build up a kind of boolean logic in the language of relations, where the empty relation is **false** and the relation containing $()$ is **true**.

Outside of mathematics, the kinds of relations normally modelled in databases are of the discrete, finite-sized variety. A relation over tuples (a, b, c) is equivalent to table with three columns a , b and c . In a *strictly* relational databases, the ordering of the rows is irrelevant, and you cannot have two identical rows in the database (though in practice many databases in use may also deal with “bags” rather than “sets” of tuples, where rows can be repeated). The fields a , b and c are drawn from their own set of possibilities, or *datatype*. For example, you might have $(\text{name}, \text{birthdate}, \text{is_adult})$ where **name** may be a string, **birthdate** may be a date, and **is_adult** may be a boolean.

name	birthdate	is_adult
"Alice"	2003-05-13	true
"Bob"	2005-12-10	false
"Charlie"	2004-09-21	true

Note that the latter two columns of this particular relation are a function if we assume **is_adult** is defined in terms of age (and thus in terms of **birthdate**).

In many relations, there are uniqueness constraints - for example the **name** might identify each row uniquely, or there may be a specific ID column to deal with different people sharing the same name. Such unique constraints are called “keys” and the primary identifier that you might use to fetch a particular row is called the “primary key”.

2.2 Relational schema

A single relation is a restrictive way to model your data. However, by allowing multiple relations, we can capture the various sets of data in the system, and piece them together by their *relationships*.

For example, each person in the table above might be related to other things. They might have jobs, bank accounts, etc. Within a single business a person might be a customer, a supplier and an employee, and for different purposes we might want data associated with that person (e.g. their birthdate might relate to advertizing promotions to customers, payrates for employees, etc).

Generally, to make keeping track of data easy, in a relational database you would store that person *just once* and create relationships between that person and other things. That way, if something changes about that person, it only needs to be updated in one place, and data remains consistent everywhere.

Here is an example of a more complex example - an imagining of how *LinkedIn* might store their data in a relational database (taken from “Designing Data-Intensive Applications” by Martin Kleppmann).

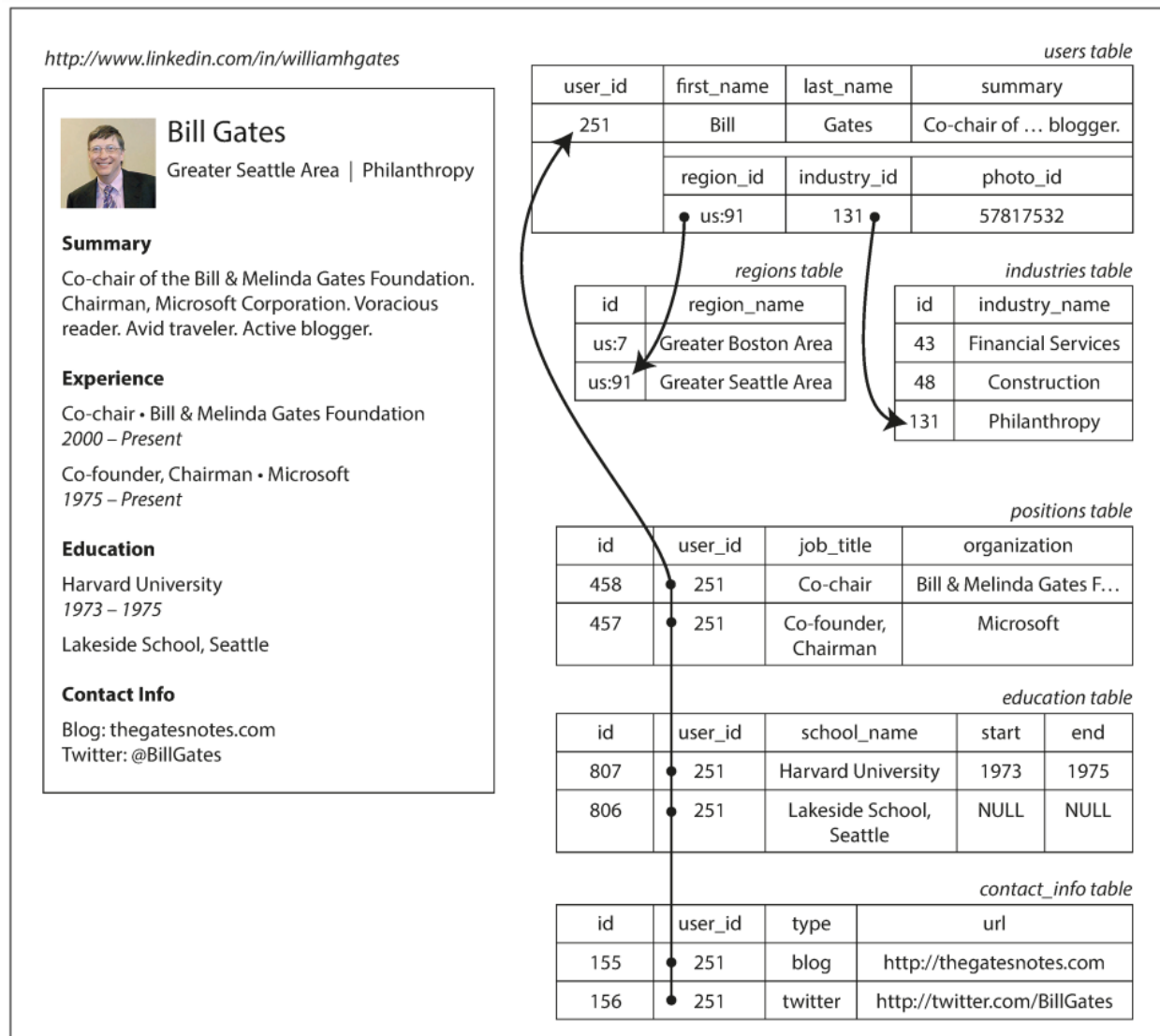


Figure 2-1. Representing a LinkedIn profile using a relational schema. Photo of Bill Gates courtesy of Wikimedia Commons, Ricardo Stuckert, Agência Brasil.

Figure 1: From *Designing Data-Intensive Applications* by Martin Kleppmann

2.3 SQL

The most popular way to interact with relational data is with via a language called SQL (originally “SE-QUEL”, pronounced that way, backronymed to “structured query language”).

The language is from 1973 and doesn’t look like most other programming languages. With it, you declare a “query” and the database will find the best way to return the results. It is a form of [declarative programming](#). Here is an example of getting some columns of data from the `users` table:

```
SELECT user_id, first_name, last_name
FROM users
```

You can filter the rows via a `WHERE` clause. Let’s say you knew the `user_id` for Bill Gates and you wanted just his data:

```
SELECT user_id, first_name, last_name
FROM users
WHERE user_id = 251
```

Data in different tables are related via the JOIN statement

```
SELECT users.user_id, first_name, last_name, organization
INNER JOIN positions ON positions.user_id = users.user_id
FROM users
```

Note we have to prefix the shared column names with the corresponding table (although in this case the difference is not particularly important, your query cannot be ambiguous).

But there is a problem here. This query would create a large amount of data, that the database would need to collect and send over the network to you. In the worst case, performing such a query could bring down a large system!

Generally, you have something more precise in mind. Like - which organizations has Bill Gates worked at?

```
SELECT users.user_id, first_name, last_name, organization
INNER JOIN positions ON positions.user_id = users.user_id
FROM users
WHERE users.user_id = 251
```

This will now return just two rows (for Microsoft, and the Bill & Melinda Gates Foundation), and is much less effort for the database, the network and the client. In real-world settings, complex joins could link data spanning dozens of tables.

There are several Julia libraries for interacting with SQL databases including the likes of [LibPQ.jl](#) allowing queries to be executed on a PostgreSQL database and the results returned as convenient Julia data structures. There's even an SQL REPL in [SQLRepl.jl](#) if you want to execute SQL queries interactively and mix them with variables from your normal Julia session.

We won't be writing SQL in this course but it is so pervasive in industry that is essential that you know that it exists and not to be afraid to use it! (It is a very useful and desirable skill!) All the relational data concepts expressed in SQL above also apply to in-memory data analysis using Julia and we show various examples of this next.

3 Dataframes

Working with tabular data is a central part of data analysis (think Excel). Data has rows (observations) and columns (variables/features). Typically a row would be for an individual, or item, or event. Typically a column would be for attributes of the individual, properties of the events, etc. Variables can be numerical, categorical, strings, or even more complex entities (e.g. images, arrays or dictionaries).

The datafile `athlete_events.csv` is a moderately sized dataset available from [Kaggle](#) covering all athletes that have participated the Summer or Winter Olympics between 1896 and 2016. The CSV file contains 40MB of data over 15 columns and more than 27,000 rows, which is “small data” for a modern computer (but would have been challenging in the 1980's). In this context, “big data” is too large to be handled by a single machine, and we'll discuss “big data” and “medium data” later.

You have already seen how to read and write from CSV files in Unit 3 where you also explored JSON files. You may recall using the `DataFrames.jl` package in that unit. We now take a deeper dive into the concept of dataframes.

We will use `athlete_events.csv` to show basic functionality of the [DataFrames.jl](#) and [TypedTables.jl](#) packages. In a sense these packages are alternatives. Seeing some functionality from each may be useful for gaining a general understanding of what to expect from such packages. If you work with Python, then using

`pandas` is the common alternative. Similarly if you work with the R language then the built-in dataframes are common.

3.1 Basic data exploration

Whenever you receive a dataset, it takes some time to understand its contents. This process is generally known as “data exploration”. To do so, we generally load (a subset of) the data from disk and do some basic analysis in a Jupyter notebook or at the REPL.

We were introduced to CSV files earlier in the course. Here’s the first few lines of our file:

```
open("../data/athlete_events.csv") do io
  i = 0
  while i < 5
    println(readline(io))
    i += 1
  end
end
```

```
SystemError: SystemError("opening file \"../data/athlete_events.csv\"", 2, nothing)
SystemError: opening file "../data/athlete_events.csv": No such file or directory
Stacktrace:
 [1] systemerror(p::String, errno::Int32; extrainfo::Nothing)
      @ Base ./error.jl:186
 [2] open(fname::String; lock::Bool, read::Nothing, write::Nothing, create::Nothing,
truncate::Nothing, append::Nothing)
      @ Base ./iostream.jl:317
 [3] open
      @ ./iostream.jl:296 [inlined]
 [4] open(f::var"#2#3", args::String; kwargs::@Kwargs{})
      @ Base ./io.jl:410
 [5] open(f::Function, args::String)
      @ Base ./io.jl:409
 [6] top-level scope
      @ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:125
```

The `do` syntax here injects a function which takes `io::IO` (an I/O stream for our file) into the first argument of the `open` function. It is syntax sugar. The `open` function is designed such that when the inner function returns or throws an error, `open` will automatically close the file to avoid resource leakage.

We can use the `CSV.jl` package to read the data and the `DataFrames.jl` package to hold the tabular data. We note in the above the appearance of "NA" for missing data, which we can use to help load the file.

```
using DataFrames, CSV
csv_file = CSV.File("../data/athlete_events.csv"; missingstring = "NA")
df = DataFrame(csv_file)
println("Data size: ", size(df))
println("Columns in the data: ", names(df))
```

```
ArgumentError: ArgumentError("\"../data/athlete_events.csv\" is not a valid file or doesn't exist")
ArgumentError: "../data/athlete_events.csv" is not a valid file or doesn't exist
Stacktrace:
 [1] macro expansion
      @ ~/.julia/packages/CSV/XLcQT/src/context.jl:314 [inlined]
 [2] Context
      @ ./none:0 [inlined]
```

```
[3] CSV.Context(source::CSV.Arg, header::CSV.Arg, normalizenames::CSV.Arg, datarow::CSV.Arg,
skipto::CSV.Arg, footerskip::CSV.Arg, transpose::CSV.Arg, comment::CSV.Arg,
ignoreemptyrows::CSV.Arg, ignoreemptylines::CSV.Arg, select::CSV.Arg, drop::CSV.Arg,
limit::CSV.Arg, buffer_in_memory::CSV.Arg, threaded::CSV.Arg, ntasks::CSV.Arg, tasks::CSV.Arg,
rows_to_check::CSV.Arg, lines_to_check::CSV.Arg, missingstrings::CSV.Arg, missingstring::CSV.Arg,
delim::CSV.Arg, ignorerepeated::CSV.Arg, quoted::CSV.Arg, quotechar::CSV.Arg,
openquotechar::CSV.Arg, closequotechar::CSV.Arg, escapechar::CSV.Arg, dateformat::CSV.Arg,
dateformats::CSV.Arg, decimal::CSV.Arg, groupmark::CSV.Arg, truemstrings::CSV.Arg,
falsestrings::CSV.Arg, stripwhitespace::CSV.Arg, type::CSV.Arg, types::CSV.Arg, typemap::CSV.Arg,
pool::CSV.Arg, downcast::CSV.Arg, lazystrings::CSV.Arg, stringtype::CSV.Arg, strict::CSV.Arg,
silencewarnings::CSV.Arg, maxwarnings::CSV.Arg, debug::CSV.Arg, parsingdebug::CSV.Arg,
validate::CSV.Arg, streaming::CSV.Arg)
@ CSV ~/.julia/packages/CSV/XLcQT/src/context.jl:0
[4] #File#18
@ ~/.julia/packages/CSV/XLcQT/src/file.jl:222 [inlined]
[5] top-level scope
@ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:140
```

Since the data is large, we don't usually want to print it all. The `first` and `last` functions can be helpful:

```
first(df, 10)
```

```
UndefVarError: UndefVarError(:df, 0x00000000000098ac, Main.Notebook)
UndefVarError: `df` not defined in `Main.Notebook`
Suggestion: check for spelling errors or missing imports.
Hint: a global variable of this name also exists in Rmath.
Stacktrace:
 [1] top-level scope
@ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:149
```

```
last(df, 10)
```

```
UndefVarError: UndefVarError(:df, 0x00000000000098ac, Main.Notebook)
UndefVarError: `df` not defined in `Main.Notebook`
Suggestion: check for spelling errors or missing imports.
Hint: a global variable of this name also exists in Rmath.
Stacktrace:
 [1] top-level scope
@ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:153
```

This dataset appears to be sorted by last name. Each row corresponds to an entrant to an event. The ID column identifies each unique athlete.

A `DataFrame` is indexed like a 2D container of rows and columns. The `:` symbol means “all”.

```
df[1:10, :]
```

```
UndefVarError: UndefVarError(:df, 0x00000000000098ac, Main.Notebook)
UndefVarError: `df` not defined in `Main.Notebook`
Suggestion: check for spelling errors or missing imports.
Hint: a global variable of this name also exists in Rmath.
Stacktrace:
 [1] top-level scope
@ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:161
```

Internally, a dataframe stores one vector of data per column. You can access any of them conveniently as “properties” with `.` syntax:

```
df.ID
```

```
UndefVarError: UndefVarError(:df, 0x00000000000098ac, Main.Notebook)
UndefVarError: `df` not defined in `Main.Notebook`
```

Suggestion: check for spelling errors or missing imports.
Hint: a global variable of this name also exists in Rmath.
Stacktrace:
[1] top-level scope
@ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:167

```
df.Height
```

UndefVarError: UndefVarError(:df, 0x00000000000098ac, Main.Notebook)
UndefVarError: `df` not defined in `Main.Notebook`
Suggestion: check for spelling errors or missing imports.
Hint: a global variable of this name also exists in Rmath.
Stacktrace:
[1] top-level scope
@ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:171

As you can see from above, some data can be missing. How many heights are missing?

```
ismissing.(df.Height) |> count # Piping: `f(x) |> g` syntax means `g(f(x))`
```

UndefVarError: UndefVarError(:df, 0x00000000000098ac, Main.Notebook)
UndefVarError: `df` not defined in `Main.Notebook`
Suggestion: check for spelling errors or missing imports.
Hint: a global variable of this name also exists in Rmath.
Stacktrace:
[1] top-level scope
@ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:177

The count function counts the number of true values.

How many unique athletes are there?

```
unique(df.ID) |> length
```

UndefVarError: UndefVarError(:df, 0x00000000000098ac, Main.Notebook)
UndefVarError: `df` not defined in `Main.Notebook`
Suggestion: check for spelling errors or missing imports.
Hint: a global variable of this name also exists in Rmath.
Stacktrace:
[1] top-level scope
@ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:185

The unique operation has extra methods defined for DataFrames. We can get the athlete data like so:

```
athlete_df = unique(df, :ID)
```

UndefVarError: UndefVarError(:df, 0x00000000000098ad, Main.Notebook)
UndefVarError: `df` not defined in `Main.Notebook`
Suggestion: check for spelling errors or missing imports.
Hint: a global variable of this name also exists in Rmath.
Stacktrace:
[1] top-level scope
@ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:191

It keeps all columns of the first row containing each distinct ID, which will we take as representative (for their Sex, Height, Weight, etc).

3.2 Simple analysis

Now we are becoming familiar with the dataset, we might like to see what insights we can gain. For example, let's see how many men and women have competed.


```
count=="M"), athlete_df.Sex) # `==(x)` creates a function `y -> y == x`
```

```
UndefinedVarError: UndefinedVarError(:athlete_df, 0x00000000000098ad, Main.Notebook)
UndefinedVarError: `athlete_df` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:
 [1] top-level scope
      @ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:201
```

```
count=="F"), athlete_df.Sex)
```

```
UndefinedVarError: UndefinedVarError(:athlete_df, 0x00000000000098ad, Main.Notebook)
UndefinedVarError: `athlete_df` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:
 [1] top-level scope
      @ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:205
```

The `count` function is useful, but this gets tiresome for many “groups” of data. The [SplitApplyCombine.jl](#) package contains useful functions for grouping data.

```
using SplitApplyCombine
```

```
groupcount(athlete_df.Sex)
```

```
UndefinedVarError: UndefinedVarError(:athlete_df, 0x00000000000098ae, Main.Notebook)
UndefinedVarError: `athlete_df` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:
 [1] top-level scope
      @ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:213
```

Which allows us to explore larger sets of groupings:

```
# How many athletes by country
team_sizes = groupcount(athlete_df.Team)
```

```
UndefinedVarError: UndefinedVarError(:athlete_df, 0x00000000000098af, Main.Notebook)
UndefinedVarError: `athlete_df` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:
 [1] top-level scope
      @ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:220
```

One of the most powerful tricks in data analysis is to sort your data.

```
sort(team_sizes)
```

```
UndefinedVarError: UndefinedVarError(:team_sizes, 0x00000000000098af, Main.Notebook)
UndefinedVarError: `team_sizes` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:
 [1] top-level scope
      @ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:225
```

Note that `groupcount` is a specialization of the more flexible `group(keys, values)` function:

```
group(athlete_df.Team, athlete_df.Name)
```

```
UndefinedVarError: UndefinedVarError(:athlete_df, 0x00000000000098af, Main.Notebook)
UndefinedVarError: `athlete_df` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
```

Stacktrace:

```
[1] top-level scope
@ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:231
```

Rather than acting at the level of vectors, you can group an entire `DataFrame` into a `GroupedDataFrame` with the `groupby` function provided by *DataFrames.jl*:

```
gdf = groupby(athlete_df, :Team)
```

```
UndefVarError: UndefVarError(:athlete_df, 0x00000000000098b0, Main.Notebook)
UndefVarError: `athlete_df` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:
```

```
[1] top-level scope
@ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:237
```

You can apply an operation to each group and bring the data together into a single `DataFrame` via the `combine` function.

```
combine(gdf, nrow => :Count)
```

```
UndefVarError: UndefVarError(:gdf, 0x00000000000098b0, Main.Notebook)
UndefVarError: `gdf` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:
```

```
[1] top-level scope
@ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:243
```

The `nrow => :Count` syntax is a convenience provided by *DataFrames.jl*. We'll explain how it works below.

3.3 Row- and column-based storage

What is a dataframe?

It is a data structure containing rows and columns. Internally it keeps each column of data as its own vector, which you can access as a property with `.`

```
athlete_df.Height
```

```
UndefVarError: UndefVarError(:athlete_df, 0x00000000000098b0, Main.Notebook)
UndefVarError: `athlete_df` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:
```

```
[1] top-level scope
@ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:255
```

To construct a row, you need to grab the corresponding element from each vector.

```
first(eachrow(df))
```

```
UndefVarError: UndefVarError(:df, 0x00000000000098b0, Main.Notebook)
UndefVarError: `df` not defined in `Main.Notebook`
Suggestion: check for spelling errors or missing imports.
Hint: a global variable of this name also exists in Rmath.
Stacktrace:
```

```
[1] top-level scope
@ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:261
```

Or, you can use 2D indexing, as before

```
df[1, :]
```

```
UndefVarError: UndefVarError(:df, 0x00000000000098b0, Main.Notebook)
```

```

UndefVarError: `df` not defined in `Main.Notebook`
Suggestion: check for spelling errors or missing imports.
Hint: a global variable of this name also exists in Rmath.
Stacktrace:

```

```

[1] top-level scope
@ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:267

```

In this case there are 15 columns, so the computer must fetch data from 15 different places. This means that, for dataframes, operating with columns is faster than with rows. DataFrames are specialized for whole-of-table analytics, where each individual step in your analysis probably only involves a small number of columns.

There are other data structures you can use which store the data as rows. Most SQL databases will store their data like this. Traditional “transactional” databases are typically driven with reads and writes to one (or a few) rows at a time, and row-based storage is more efficient for such workloads. Some of the modern “analytics” databases will use column-based storage.

In Julia, perhaps the simplest such data structure is a `Vector` of `NamedTuples`. We can create one straightforwardly from a `CSV.File`:

```
NamedTuple.(csv_file)
```

```

UndefVarError: UndefVarError(:csv_file, 0x00000000000098b0, Main.Notebook)
UndefVarError: `csv_file` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:
[1] top-level scope
@ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:277

```

The above constructs a `NamedTuple` for each row of the CSV file and and returns it as a `Vector`.

This is a great data structure that you can use “out of the box” with no packages, and your “everyday” analysis work will usually be fast so long as you do not have many, many columns.

If you want to use an identical interface with row-based storage, there is the [TypedTables](#) package. In this package, a `Table` is an `AbstractArray{<:NamedTuple}`, each column is stored as its own vector, and when you index the table `table[i]` it assembles the row as a `NamedTuple` for you.

```

using TypedTables
Table(csv_file)

```

```

UndefVarError: UndefVarError(:csv_file, 0x00000000000098b1, Main.Notebook)
UndefVarError: `csv_file` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:
[1] top-level scope
@ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:288

```

Sometimes plain vectors or (typed) tables may be more convenient or faster than dataframes, and sometimes dataframes will be more convenient and faster than plain vectors or typed tables. Another popular approach is to use *Query.jl*. For your assessment you may use whichever approach you like best.

3.4 DataFrames

For now we will double-down on `DataFrames.jl`.

Here is a [nice cheatsheet for the syntax of DataFrames](#), and I recommend downloading and possibly even printing it out for your convenience.

3.4.1 Constructing dataframes

A `DataFrame` can be constructed directly from data columns.

```
df = DataFrame(a = [1, 2, 3], b = [2.0, 4.0, 6.0])
```

	a	b
	Int64	Float64
1	1	2.0
2	2	4.0
3	3	6.0

You can get or set columns to the dataframe with `.` property syntax.

```
df.a
```

```
3-element Vector{Int64}:
```

```
1
2
3
```

```
df.c = ["A", "B", "C"]
df
```

	a	b	c
	Int64	Float64	String
1	1	2.0	A
2	2	4.0	B
3	3	6.0	C

3.4.2 Indexing dataframes

A dataframe is indexed a bit like a 2D matrix - the first index is the row and the second is the column.

```
df[1, :c]
```

```
"A"
```

The `:c` here is a `Symbol`, which is a kind of “compiler string”. The compiler stores the names of your types, fields, variables, modules and functions as `Symbol`. In fact, the syntax `df.c` is just sugar for `getproperty(df, :c)`. You can get multiple rows and/or columns.

```
df[1, :]
```

	a	b	c
	Int64	Float64	String
1	1	2.0	A

```
df[:, :c]
```

```
3-element Vector{String}:
```

```
"A"
"B"
"C"
```

```
df[1:2, [:a, :c]]
```

	a	c
	Int64	String
1	1	A
2	2	B

3.4.3 Filtering dataframes

The `filter` function returns a collection where the elements satisfy some predicate function (i.e. a function that returns a `Bool`). We can grab just the odd-numbered rows from `df` by running `filter` over each row.

```
filter(row -> isodd(row.a), df)
```

	a	b	c
	Int64	Float64	String
1	1	2.0	A
2	3	6.0	C

Unfortunately, in this case this is quite inefficient. For each row the program must

1. construct a `DataFrameRow`
2. access the `a` field *dynamically*
3. compute `isodd`

This is slower than we'd like because the compiler doesn't really know what columns are in a `DataFrame` (or a `DataFrameRow`) and what type they may be. Every time we do step 2 there is a lot of overhead.

To fix this problem *DataFrames.jl* introduced special syntax of the form:

```
filter(:a => isodd, df)
```

	a	b	c
	Int64	Float64	String
1	1	2.0	A
2	3	6.0	C

This will automatically extract the `:a` column *just once* and use it to construct a fast & fully compiled predicate function. Another way to think about how this works is via indexing:

```
df[isodd.(df.a), :]
```

	a	b	c
	Int64	Float64	String
1	1	2.0	A
2	3	6.0	C

So first we find:

```
isodd.(df.a)
```

3-element BitVector:

```
1  
0  
1
```

And afterwards we take a subset of the rows.

```
df[[1,3], :]
```

	a	b	c
	Int64	Float64	String
1	1	2.0	A
2	3	6.0	C

3.4.4 select and transform

You can grab one-or-more columns via `select`:

```
select(df, [:a, :c])
```

	a	c
	Int64	String
1	1	A
2	2	B
3	3	C

The columns can be modified or even renamed as a part of this process. This is more useful with **transform**, which keeps the existing columns and lets you add new ones:

```
using Statistics
transform(df, :b => mean => :mean_b)
```

	a	b	c	mean_b
	Int64	Float64	String	Float64
1	1	2.0	A	4.0
2	2	4.0	B	4.0
3	3	6.0	C	4.0

You can read this as “take column **b**, do **mean** on it, and put the result in column **mean_b**”. A new dataframe is constructed, and **df** is unmodified.

Note that the transformation applies to whole columns. If you want to transform just a single row at a time, wrap the function in a **ByRow**.

```
transform!(df, :a => ByRow(isodd) => :a_isodd, :c => ByRow(lowercase) => :c_lowercase)
```

	a	b	c	a_isodd	c_lowercase
	Int64	Float64	String	Bool	String
1	1	2.0	A	1	a
2	2	4.0	B	0	b
3	3	6.0	C	1	c

Here we used **transform!** (note the **!**) which mutates **df**.

3.4.5 groupby and combine

The **groupby** function will group a **DataFrame** into a collection of **SubDataFrames**:

```
gdf = groupby(df, :a_isodd)
```

GroupedDataFrame with 2 groups based on key: **a_isodd**

First Group (1 row): **a_isodd** = false

	a	b	c	a_isodd	c_lowercase
	Int64	Float64	String	Bool	String
1	2	4.0	B	0	b

...

Last Group (2 rows): **a_isodd** = true

	a	b	c	a_isodd	c_lowercase
	Int64	Float64	String	Bool	String
1	1	2.0	A	1	a
2	3	6.0	C	1	c

You can combine these together by applying a bulk transformation to each group

```
combine(gdf, :b => sum, :c => join)
```

	a_isodd	b_sum	c_join
	Bool	Float64	String
1	0	4.0	B
2	1	8.0	AC

This is known as the split-apply-combine strategy, and the pattern comes up frequently.

3.4.6 innerjoin

We won't use this a lot in this course, but you can perform a relational join between dataframes with the `innerjoin` function. (Note that the `join` function is for joining strings together into longer strings)

```
names_df = DataFrame(ID = [1, 2, 3], Name = ["John Doe", "Jane Doe", "Joe Blogs"])
```

	ID	Name
	Int64	String
1	1	John Doe
2	2	Jane Doe
3	3	Joe Blogs

```
jobs = DataFrame(ID = [1, 2, 4], Job = ["Lawyer", "Doctor", "Farmer"])
```

	ID	Job
	Int64	String
1	1	Lawyer
2	2	Doctor
3	4	Farmer

```
DataFrames.innerjoin(names_df, jobs; on = :ID)
```

	ID	Name	Job
	Int64	String	String
1	1	John Doe	Lawyer
2	2	Jane Doe	Doctor

Only rows with matching `:IDs` are kept.

3.5 More analysis

Now that we know a little bit more about the tools, let's use them to see what insights we can glean from our Olympic athlete data.

3.5.1 Athlete height

We can see that, on average, male competitors are taller than female competitors.

```
mean(athlete_df.Height[athlete_df.Sex .== "M"])
```

```

UndefVarError: UndefVarError(:athlete_df, 0x00000000000098b8, Main.Notebook)
UndefVarError: `athlete_df` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:
 [1] top-level scope
      @ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:450

```

Oops. Not all athletes have a `Height` attribute.

```
mean(skipmissing(athlete_df.Height[athlete_df.Sex .== "M"]))
```

```
UndefVarError: UndefVarError(:athlete_df, 0x00000000000098b8, Main.Notebook)
UndefVarError: `athlete_df` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:
 [1] top-level scope
      @ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:456
```

```
mean(skipmissing(athlete_df.Height[athlete_df.Sex .== "F"]))
```

```
UndefVarError: UndefVarError(:athlete_df, 0x00000000000098b8, Main.Notebook)
UndefVarError: `athlete_df` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:
 [1] top-level scope
      @ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:460
```

The males are a bit more than 10cm taller, on average.

OK, now let's perform a slightly more complex analysis. We will answer the question - has athlete height has changed over time? What do you think?

We can plot the average height as a function of Year. To see how to do that, first we'll repeat the above using the tools of *DataFrames.jl*:

```
athlete_by_gender = groupby(athlete_df, :Sex)
combine(athlete_by_gender, :Height => mean skipmissing => :Height)
```

```
UndefVarError: UndefVarError(:athlete_df, 0x00000000000098b9, Main.Notebook)
UndefVarError: `athlete_df` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:
 [1] top-level scope
      @ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:470
```

Given that, it's pretty straightforward to do this as a function of year.

```
using Plots
athlete_by_year = groupby(athlete_df, :Year)
height_by_year = combine(athlete_by_year, :Height => mean skipmissing => :Height)
plot(height_by_year.Year, height_by_year.Height, ylim = [155, 182]; ylabel = "Height (cm)", xlabel = "Year", legend = false)
```

```
UndefVarError: UndefVarError(:athlete_df, 0x00000000000098bb, Main.Notebook)
UndefVarError: `athlete_df` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:
 [1] top-level scope
      @ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:478
```

This doesn't show anything interesting. Yet. There are a few confounding factors to eliminate first.

First, there is something going on strange with the years, which started at 4-year increments and changed to 2-year increments. This is due to Winter Olympics moving to a different year to the Summer Olympics in 1994.

```
athlete_by_games = groupby(athlete_df, [:Year, :Season])
height_by_games = combine(athlete_by_games, :Height => mean skipmissing => :Height)
plot(height_by_games.Year, height_by_games.Height, ylim = [155, 182]; ylabel = "Height (cm)",
xlabel = "Year", group = height_by_games.Season, legend = :bottomright)
```



```

UndefVarError: UndefVarError(:athlete_df, 0x00000000000098bc, Main.Notebook)
UndefVarError: `athlete_df` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:
 [1] top-level scope
      @ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:488

```

The type of sport might affect the height of the competitors (think basketball players vs jockeys) so it's good to split these groups. Here it seems that winter competitors are slightly shorter than summer competitors.

The second confounding factor is that women have increasingly become a larger fraction of the competitors at the Olympics, so we will split also by gender.

```

athlete_by_cohort = groupby(athlete_df, [:Year, :Season, :Sex])
height_by_cohort = combine(athlete_by_cohort, :Height => mean skipmissing => :Height)
plot(height_by_cohort.Year, height_by_cohort.Height, ylim = [155, 182]; ylabel = "Height (cm)",
      xlabel = "Year", group = tuple.(height_by_cohort.Season, height_by_cohort.Sex), legend =
      :bottomright)

```

```

UndefVarError: UndefVarError(:athlete_df, 0x00000000000098bd, Main.Notebook)
UndefVarError: `athlete_df` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:
 [1] top-level scope
      @ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:498

```

Whoa! Now we clearly see that heights have trended up at least 5cm since 1900!

What's going on here? I suspect it is a combination of several facts:

1. On average, people born before the end of World War 2 were shorter, due to nutritional changes.
2. Sport has become more elite and competitive over the years, and height may correlate with success in many Olympic sports.
3. Women competitors have become more prevalent over time, reducing the average height of *all* competitors.
4. Similarly, winter competitors may have become more prevalent over time, who appear to be shorter than the summer cohorts.

We can easily verify #3 above.

```

gender_by_games = combine(athlete_by_games, :Sex => (s -> count(==( "F"), s) / length(s)) =>
:Fraction_Female)
plot(gender_by_games.Year, gender_by_games.Fraction_Female, ylim = [0, 1]; ylabel = "Fraction
female", xlabel = "Year", group = gender_by_games.Season, legend = :topright)

```

```

UndefVarError: UndefVarError(:athlete_by_games, 0x00000000000098c1, Main.Notebook)
UndefVarError: `athlete_by_games` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:
 [1] top-level scope
      @ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:515

```

Note that we could have given up our analysis with the first plot, above, and arrived at completely the wrong conclusion!

Having these tools under your belt is a very useful skill. It is quite common that you need to have the skills to dig under the surface to get a correct understanding of data. It is just as useful to debunk a myth or assumption as it is to find a hitherto unknown correlation.

3.5.2 Histograms

These are just the means, we can also compare the statistic distribution (for the 2016 games, say):

```

histogram(collect(skipmissing(athlete_by_cohort[(2016, "Summer", "M")].Height)), xlabel = "Height
(cm)", opacity = 0.5, label = "Male")
histogram!(collect(skipmissing(athlete_by_cohort[(2016, "Summer", "F")].Height)), opacity = 0.5,
label = "Female")

```

```

UndefVarError: UndefVarError(:athlete_by_cohort, 0x000000000000098c1, Main.Notebook)
UndefVarError: `athlete_by_cohort` not defined in `Main.Notebook`
Suggestion: add an appropriate import or assignment. This global was declared but not assigned.
Stacktrace:

```

```

[1] top-level scope
@ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:528

```

The `histogram` function is useful for identifying features of distributions (e.g. if it is bimodal).

3.5.3 Further questions

We could analyse this data another hundred ways. Some questions that come to mind are:

- How does team success relate to socioeconomic indicators of their home country, such as GDP per capita? Do richer countries do comparatively better than poorer countries? To do this, we would need to *join* the data with country data.
- Does team success depend on the distance between the host of the Olympics and the home nation? For example, Australia received a lot of medals during the Sydney 2000 Olympics.
- Can we predict how well a team will do at a given Olympics, based on data like above? This is heading in the direction of machine learning.

3.6 Fitting linear models

Here we'll briefly show how to fit linear models to your data using the package **GLM.jl**. A one dimensional linear model may be written

$$Y = \beta_0 + \beta_1 X + \epsilon$$

where Y is some dependent variable to be predicted, X is the independent variable and ϵ is some random noise. The task of fitting is to estimate the model parameters β_0 and β_1 ; one may also want to estimate the variance of the noise ϵ . Simple linear models like this (and the multidimensional generalization) can be fit with least squares to give the maximum likelihood estimate of the parameters β .

Let's look at a simple example from the athletes data

```

using GLM, Underscores

df = DataFrame(CSV.File("../data/athlete_events.csv"; missingstring = "NA"))

rowing_by_year = @_ df |>
  groupby(__, [:Year, :Sport]) |>
  combine(__, nrow => :Count) |>
  sort(__, :Year) |>
  filter(:Sport => ==("Rowing"), __)

model = lm(@formula(Count ~ Year), rowing_by_year)

scatter(rowing_by_year.Year, rowing_by_year.Count, label="Data")
plot!(rowing_by_year.Year, predict(model, rowing_by_year), label="Prediction")

```

```

ArgumentError: ArgumentError("\n../data/athlete_events.csv\n" is not a valid file or doesn't exist")
ArgumentError: "../data/athlete_events.csv" is not a valid file or doesn't exist
Stacktrace:

```

```

[1] macro expansion
   @ ~/.julia/packages/CSV/XLcqT/src/context.jl:314 [inlined]
[2] Context
   @ ./none:0 [inlined]
[3] CSV.Context(source::CSV.Arg, header::CSV.Arg, normalizenames::CSV.Arg, datarow::CSV.Arg,
skipto::CSV.Arg, footerskip::CSV.Arg, transpose::CSV.Arg, comment::CSV.Arg,
ignoreemptyrows::CSV.Arg, ignoreemptylines::CSV.Arg, select::CSV.Arg, drop::CSV.Arg,
limit::CSV.Arg, buffer_in_memory::CSV.Arg, threaded::CSV.Arg, ntasks::CSV.Arg, tasks::CSV.Arg,
rows_to_check::CSV.Arg, lines_to_check::CSV.Arg, missingstrings::CSV.Arg, missingstring::CSV.Arg,
delim::CSV.Arg, ignorerepeated::CSV.Arg, quoted::CSV.Arg, quotechar::CSV.Arg,
openquotechar::CSV.Arg, closequotechar::CSV.Arg, escapechar::CSV.Arg, dateformat::CSV.Arg,
dateformats::CSV.Arg, decimal::CSV.Arg, groupmark::CSV.Arg, truemstrings::CSV.Arg,
falsestrings::CSV.Arg, stripwhitespace::CSV.Arg, type::CSV.Arg, types::CSV.Arg, typemap::CSV.Arg,
pool::CSV.Arg, downcast::CSV.Arg, lazystrings::CSV.Arg, stringtype::CSV.Arg, strict::CSV.Arg,
silencewarnings::CSV.Arg, maxwarnings::CSV.Arg, debug::CSV.Arg, parsingdebug::CSV.Arg,
validate::CSV.Arg, streaming::CSV.Arg)
   @ CSV ~/.julia/packages/CSV/XLcqT/src/context.jl:0
[4] #File#18
   @ ~/.julia/packages/CSV/XLcqT/src/file.jl:222 [inlined]
[5] top-level scope
   @ ~/nextcloud/MATH2504_2026/lecture/quarto/lecture-unit-8.qmd:563

```

Q: Is this linear fit a reasonable thing to do? (What's going on after 2000?)

Let's now look at how to fit a multidimensional model using GLM.jl. In the multidimensional case we have several independent variables $X_1, \dots, X_p$ and several dependent variables $Y_1, \dots, Y_n$:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots \beta_p X_{ip} + \epsilon_i$$

Here's an example of fitting CPU performance for some historical CPUs, taken from the book "Statistics with Julia":

```

using RDatasets, Statistics

df = dataset("MASS", "cpus")
df.Freq = map( x->10^9/x , df.CycT)

model = lm(@formula(Perf ~ MMax + Cach + ChMax + Freq), df)
println("n = ", size(df)[1])
println("(Avg,Std) of observed performance: ", (mean(df.Perf),std(df.Perf)))
model

```

n = 209

(Avg,Std) of observed performance: (105.61722488038278, 160.83058719907774)

StatsModels.TableRegressionModel{LinearModel{GLM.LmResp{Vector{Float64}}, GLM.DensePredChol{Float64, Lin

Perf ~ 1 + MMax + Cach + ChMax + Freq

Coefficients:

	Coef.	Std. Error	t	Pr(> t)	Lower 95%	Upper 95%
(Intercept)	-46.5763	7.62382	-6.11	<1e-08	-61.6078	-31.5447
MMax	0.00841457	0.000639214	13.16	<1e-28	0.00715426	0.00967489
Cach	0.872508	0.152825	5.71	<1e-07	0.571189	1.17383
ChMax	0.96736	0.234847	4.12	<1e-04	0.504321	1.4304
Freq	9.74951e-7	5.5502e-7	1.76	0.0805	-1.1936e-7	2.06926e-6

Using the model to predict:

```
pred(x) = round(coef(model)'*vcat(1,x), digits = 3)
println("Estimated performance test computers: for computer A: ", pred([32000, 32, 32, 4*10^7]))

# Using the data and the `predict()` function
test_data = DataFrame(
  name = ["Computer A", "Computer B"],
  MMax = [32000, 32000],
  Cach = [32, 16],
  ChMax = [32, 32],
  Freq = [4*10^7, 6*10^7],
)

println("Estimated performance test computers:")
println(predict(model, test_data))
```

```
Estimated performance test computers: for computer A: 320.564
Estimated performance test computers:
Union{Missing, Float64}[320.56388734263396, 326.10276642540515]
```

Note that GLM stands for “generalized linear model” and the GLM.jl package can fit more general models - specifically those where the left and right hand sides are related by a nonlinear “link function”, and where the random noise is not normally distributed.

4 Memory management in data workflows

Generally in data heavy workflows we have three rough “sizes” to worry about.

1. Small data: data fits in RAM. Just load it up and process it all at once. We are doing this in this course.
2. Medium data: data is too big for RAM, but it fits on disk. Incrementally load “chunks” from disk, save the results, and load the next chunk, etc - known as “out-of-core” processing. Multiple such steps might be required in a “pipeline”. Reading and writing to disk is slower than RAM, and processing might take so long that restarting from scratch (after a fault or power blackout, for example) is not realistic, so you generally need to be able to save and restore your intermediate state, or update your data incrementally. A typical SQL database is in this category, handling the persistence, fault-tolerance and pipelining for you automatically.
3. Big data: data is too big to fit on the hard drive of a single computer. Generally requires a distributed solution for both storage and processing. At this scale, it is “business as usual” for computers to die and hard drives or RAM to corrupt, and you can’t have your multi-million-dollar operation brought down by a single rogue bitflip. These days it is common and convenient to use cloud services for these situations - like those provided by Amazon (AWS), Microsoft (Azure) or Google (GCP).

The boundaries between these regimes depends on your computer hardware. As you get step up, the complexity increases rapidly, especially with respect to fault tolerance.

At my previous job with [Fugro](#) we processed petabytes of LiDAR, imagery and positioning data to create a 3D model of the world. We used AWS to store and index the data, as well as process it. The effort in “data engineering” was as much as was required in the “data science”. Generally, it pays to have your data sorted out before attempting any higher-order analytics at scale (such as machine learning, below).